Gnocchi's GC bias is introduced by its regional adjustment, and that adjustment is fit on the wrong population

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Five panels, one argument.

	Claim	Quantity
A	The bias is *introduced by* the regional adjustment, not inherited from the context-only model	mean standardized rank vs GC, for $r \equiv 1$ and for full Gnocchi
B	That adjustment's GC dependence is wholly non-CpG — and $R_{\mathrm{CpG}} \approx 1$ is *correct*, because methylation already carries it in step 1	$R_{\mathrm{eff}} = \sum E_2 / \sum E_1$ per GC bin, decomposed by CpG status
C	The DNM training set is not the scored population: at high GC it is mostly coding, or sequence dropped for failing gnomAD's variant-call QC	composition of the training sites per GC bin
D	Restricting the training set to the scored population flattens the empirical DNM rate — and the fit then tracks it	fitted and empirical $P(\mathrm{DNM})$, non-CpG, three training populations
E	Refitting r on the scored population removes the bias in Gnocchi itself	panel A's statistic, with the retrained Gnocchi added

A and E are the same statistic on the same windows, so they are read as before/after. B, C and D explain what happens in between.

1 Where the model comes from, and where bias can enter

Expected SNVs under neutrality. In a selectively neutral window, the expected number of polymorphic sites — a site being polymorphic if an ALT allele is seen in one or more members of the cohort — is

$$\sum_i p_{c_i}(x_i),$$

where i indexes single-nucleotide sites, x_i is a vector of regional features (GC content, replication timing, …) at site i, c_i is its sequence and methylation context, and $p_c(x)$ is the probability that a site of context c with features x is polymorphic — which of course also depends on cohort size.

Uniformity within a window. Assume the feature vectors of all sites in a window w are approximately equal, $x_i = x_w$. Then

$$\sum_i p_{c_i}(x_w) \;=\; \sum_c n_c^{(w)}\, p_c(x_w),$$

grouping the left-hand terms by context, with $n_c^{(w)}$ the number of sites in w of context c.

The naive way to fit p_c. The natural approach is a parametric model

$$p_c(x) = \sigma(\beta_c \cdot x)$$

with σ the logistic function, trained per context c on a genome-wide set of sites that are monomorphic or polymorphic in the cohort.

Why that fails: the training set is contaminated by selection, and the contamination correlates with x. Those training sites ought to be selectively neutral, but we do not know how to identify neutral sites in the noncoding genome, where functional annotation is sparse and imprecise. Worse, the sites under selection *correlate with the features* — promoters are under selection and are GC-rich — so $p_c(x)$ comes out inaccurate precisely in the tails of the x distribution.

Chen et al.'s two-stage fix. De novo mutations have by definition not been exposed to selection, so they report inherent mutability. They cannot be used directly — we want a polymorphism probability, which scales with cohort size, not a DNM rate, which does not — but they can be brought in as a *ratio*. Factorize:

$$p_c(x) = r_c(x)\, p_c(\bar{x}), \qquad r_c(x) = \frac{p_c(x)}{p_c(\bar{x})},$$

with $\bar{x}$ the average of x over the windows of interest, so r_c is a feature-dependent multiplicative adjustment to a feature-independent baseline.

Step 1 estimates $p_c(\bar{x})$ by maximum likelihood over the noncoding genome,

$$p_c(\bar{x}) \;\approx\; \frac{N_c^{(p)}}{N_c^{(m)} + N_c^{(p)}},$$

with $N_c^{(m)}, N_c^{(p)}$ the numbers of monomorphic and polymorphic sites of context c. With that many training examples, most of them near $\bar{x}$, this is accurate — subject to low levels of contamination by selection.

Step 2 assumes the polymorphism probability is approximately proportional to the DNM probability $\hat{p}_c(x)$ in a training set of DNMs and matched background sites, so that

$$r_c(x) \;\approx\; \frac{\hat{p}_c(x)}{\hat{p}_c(\bar{x})}, \qquad \hat{p}_c(x) = \sigma(\hat{\beta}_c \cdot x).$$

Two consequences follow immediately, and both matter later. $\hat{p}_c$ is sensitive to the class balance of the training set (the DNM-to-background ratio), but r_c, being a ratio of two such probabilities, **is not** — this is why panel D's levels are not comparable across populations while panel B's R is. And although the DNM training set can be trusted to be neutral even in the tails, it is *far* smaller than the set behind $p_c(\bar{x})$, so r_c is the noisier of the two factors by construction.

Two places the implementation departs from this derivation, both established in `preconditions/` and both load-bearing below:

1. r is fit **per trinucleotide context only**, not per $c = (\text{trinucleotide}, \text{methylation})$. Methylation enters in step 1 alone, through $p_c(\bar{x})$. That is why $R_{\text{CpG}} \approx 1$ in panel B is *correct* rather than a failure — the large methylation effect has already been applied upstream.
2. The published Methods state r as a ratio of **logits**; the code computes the ratio of **probabilities** written above. The code is what produced the published scores (`preconditions/verify_logit_predict_behavior.py`).

2 Notation

Index 1 kb windows by w. Chen et al.'s model is a product of two factors. Write $c = (t, m)$ for a *class* of possible SNV: trinucleotide context t (32 of them) and methylation level m. Let $n_c(w)$ be the number of gnomAD-callable sites of class c in window w, and p_c the fitted per-site mutation probability for that class (`fitted_po`, keyed by $(t, \text{ref}, \text{alt}, m)$).

Step 1 — sequence context and methylation only.

$$E_1(w) \;=\; \sum_c n_c(w)\, p_c.$$

Step 2 — the regional adjustment. For each context t a logistic regression is fit on de novo mutations, then

$$r_t(w) \;=\; \frac{\sigma\!\left(\beta_{t0} + \beta_t^\top \mathbf{z}_t(w)\right)}{\sigma(\beta_{t0})}, \qquad E_2(w) \;=\; \sum_c n_c(w)\, p_c\, r_{t(c)}(w),$$

where $\mathbf{z}_t(w)$ is the standardized, PCA-transformed vector of regional features at w and σ is the logistic function. Two consequences are used throughout:

1. r is a **ratio of predicted probabilities**, so a *level* error in the fitted model — one common to numerator and denominator — cancels exactly and never reaches Gnocchi. Only feature-driven *variation* survives.
2. r_t does **not** depend on methylation: there is one model per trinucleotide context, pooling all 16 methylation levels.

(The paper's Methods state $r = \beta \cdot \mathbf{x}(w) \,/\, \beta \cdot \bar{\mathbf{x}}$, a ratio of *logits*. The code computes the ratio of probabilities above, and the code is what produced the published scores.)

The score. With $O(w)$ the observed count of rare variants,

$$\chi^2_M(w) = \frac{\big(O(w) - E_M(w)\big)^2}{E_M(w)}, \qquad z_M(w) = -\operatorname{sign}(O(w) - E_M(w))\sqrt{\chi^2_M(w)}$$

for $M \in \{1, 2\}$, keeping $|z| \leq 10$ — so high z means constrained. Standardizing to a rank within the analyzed window set,

$$\rho_M(w) = \frac{\operatorname{rank}\left(z_M(w)\right) - \frac{1}{2}}{n},$$

which is uniform on $(0, 1)$ by construction. **Panels A and E plot** $\bar{\rho}_M(g) = \mathbb{E}\big[\rho_M(w) \,\big|\, w \in g\big]$ for GC bins g. An unbiased metric sits at 0.5 in every bin; departure from 0.5 *is* the bias.

3 Panel A — the bias is introduced by the regional adjustment

Both curves are computed on one window population, with the two z statistics filtered jointly and ranked *after* that filter, so they describe exactly the same windows. The only difference between them is whether r was applied.

Depletion rank is an independently constructed metric on **Halldorsson's own windows**, so it is ranked within its own set and overlaid, never joined on `element_id`. That is legitimate here precisely because ρ is uniform on $(0, 1)$ for every curve: what is compared is how each metric's uniform mass redistributes across GC. Two consequences the caption must carry: it is binned on **its own GC edges** (its windows span a wider GC range than the Gnocchi set), and it is drawn **without error bars**, because those windows overlap — 30.4M of them after the enhancer filter, over a 3.1 Gb genome — so within-bin windows are not independent and $\mathrm{std}/\sqrt{n}$ would understate the uncertainty by about $\sqrt{\mathrm{length/step}}$. The mean curve is unaffected, and it is the only thing read off this curve.

Summary statistic quoted in the text: $\overline{|\bar{\rho}_M(g) - 0.5|}$ across bins.

Why this statistic is so sensitive to r — worth recording, because it is what licenses reading panels B–D as the *cause* of panel A. Suppose a window's adjustment is a uniform inflation, $E_2 = fE_1$, and write the observed count as $O = E_1(1 + \epsilon)$. Then

$$z_2 - z_1 = \sqrt{E_1}\left[\frac{f - 1 - \epsilon}{\sqrt{f}} + \epsilon\right] \;\approx\; \sqrt{E_1}\,(f - 1) \qquad (f \to 1,\; \epsilon \to 0).$$

The $\sqrt{E_1}$ prefactor is the point. E_1 averages 170 per window (median 181), so $\sqrt{E_1} \approx 13$, and a mere 10% inflation displaces z by more than one unit — comparable to the spread of z itself. Measured directly by re-ranking the real genome-wide z distribution under a uniform f, the mean rank moves from 0.500 at $f = 1$ to 0.705 at $f = 1.10$ and 0.284 at $f = 0.90$. Small multiplicative errors in r are not a second-order concern; they are the whole effect. (Both of those are measured on the 1,843,559-window reproduction, where the whole z distribution is available offline. They bound the sensitivity rather than reporting a result, and the narrowed set is a subset of the same windows.)

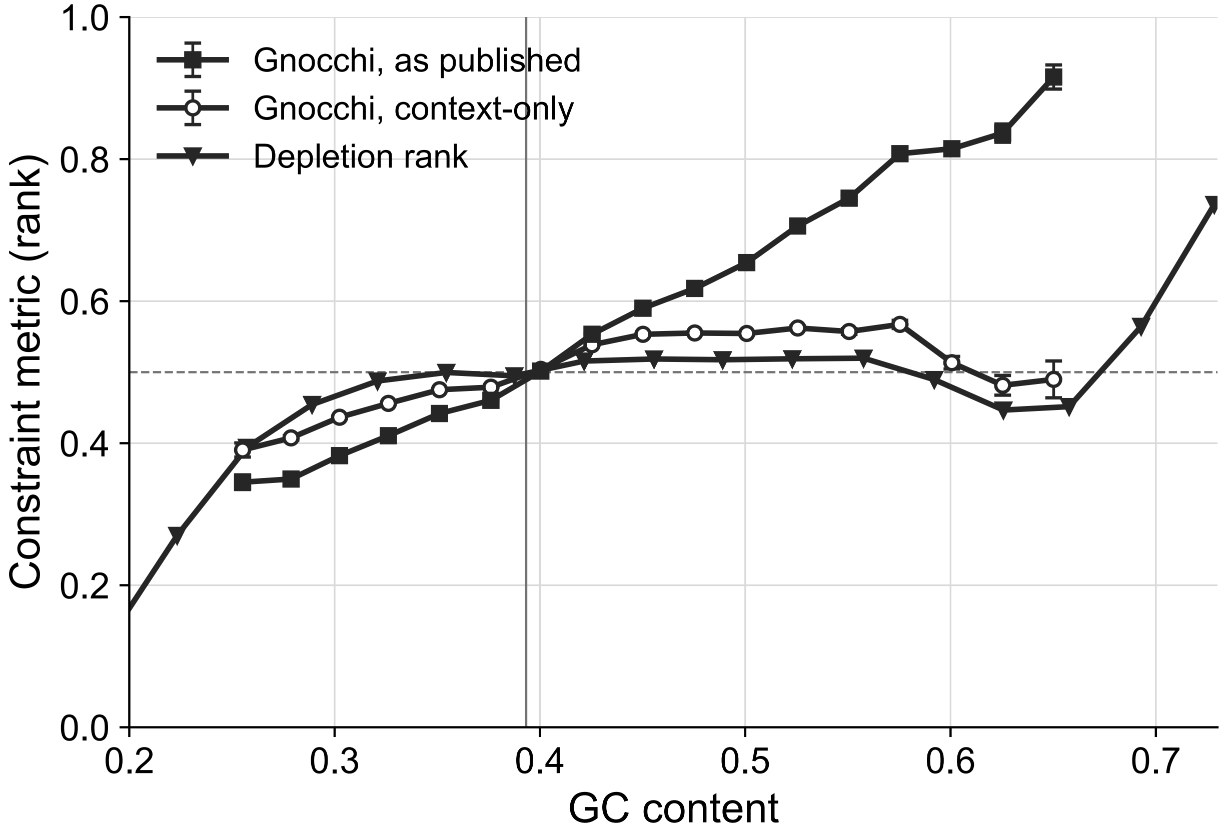


Fig. 5A | The bias is introduced by step 2, not inherited from step 1. Mean standardised rank of each window's constraint score per GC bin, under step 1 alone and under the full published model, on the same 692,348 windows filtered and ranked jointly so that the only difference between the curves is whether r was applied. High rank means constrained, and an unbiased metric sits at 0.5 in every bin. The bias is 0.046 for the context-only model and 0.168 for Gnocchi as published, which reaches a mean rank of 0.92 in its highest GC bin: high-GC windows are systematically scored as constrained. Depletion rank, an independently constructed metric on its own overlapping windows, is overlaid for comparison (0.096); it is ranked within its own set, binned on its own GC edges, and drawn without error bars.

4 Panel B — the adjustment Gnocchi applies, decomposed by CpG status

Per-context expected counts. Group step 1's sum by trinucleotide context t — the level r is fit at — writing $c \to t$ for the classes $c = (t, m)$ sharing context t:

$$E_1^t(w) = \sum_{c \to t} n_c(w)\, p_c, \qquad E_1(w) = \sum_t E_1^t(w).$$

Because $r_t(w)$ carries no methylation index, it comes out of the inner sum, and step 2 is a *reweighting of the same per-context counts*:

$$E_2^t(w) = \sum_{c \to t} n_c(w)\, p_c\, r_t(w) = E_1^t(w)\, r_t(w), \qquad E_2(w) = \sum_t E_1^t(w)\, r_t(w).$$

What one window receives. Dividing,

$$r_{\text{eff}}(w) = \frac{E_2(w)}{E_1(w)} = \sum_t \omega_t(w)\, r_t(w), \qquad \omega_t(w) = \frac{E_1^t(w)}{E_1(w)}, \quad \sum_t \omega_t(w) = 1,$$

an E_1-weighted mean of the per-context r. This is why r_{eff}, not any single r_t, is the quantity the score sees: two windows with identical r_t still receive different adjustments if their context composition differs — and context composition is precisely what varies with GC.

What a GC bin receives. Aggregate a bin g as a **ratio of summed expected counts**:

$$R_{\text{eff}}(g) = \frac{\sum_{w \in g} E_2(w)}{\sum_{w \in g} E_1(w)} = \sum_{w \in g} W(w)\, r_{\text{eff}}(w), \qquad W(w) = \frac{E_1(w)}{\sum_{w' \in g} E_1(w')},$$

again an E_1-weighted mean, of the window-level r_{eff} this time. Expected counts add, so this is the adjustment the bin actually receives, and the weights are not a separate choice — they fall out of dividing one sum by another. An unweighted mean of $r_{\text{eff}}(w)$ answers a different question, and would break the CpG decomposition below, which is an identity only for ratios of sums: bin by bin it reproduces R_{eff} to floating point ($\leq 2 \times 10^{-16}$), where the unweighted version misses by up to 4.9×10^{-3} because a window's CpG share and its non-CpG adjustment are correlated within a bin. The two aggregations agree closely here in any case: Chen et al.'s QC filter admits only windows with $\geq 1{,}000$ possible variants, so E_1 spans just 55.5–345.7, and switching to the unweighted mean would move R_{eff} by ~0.1% through the GC bulk and 0.79% at worst, in the sparsest low-GC bin. (That comparison is on the 1,843,559-window reproduction; the narrowed set is a subset of the same windows, so its E_1 span can only be tighter and the two aggregations can only agree more closely.) **Case carries the level throughout this notebook:** lowercase r is per context or per window, capital R is the bin-level aggregate, a ratio of summed expected counts.

Splitting at CpG. Let $\mathcal{K} = \{\text{ACG}, \text{CCG}, \text{GCG}, \text{TCG}\}$ and write $E_M^{\mathcal{K}} = \sum_{t \in \mathcal{K}} E_M^t$, $E_M^{\neg\mathcal{K}} = E_M - E_M^{\mathcal{K}}$ for $M \in \{1, 2\}$. Define each part's own adjustment, and the CpG share of the step-1 counts:

$$R_{\text{CpG}}(g) = \frac{\sum_{w \in g} E_2^{\mathcal{K}}(w)}{\sum_{w \in g} E_1^{\mathcal{K}}(w)}, \qquad R_{\text{non}}(g) = \frac{\sum_{w \in g} E_2^{\neg\mathcal{K}}(w)}{\sum_{w \in g} E_1^{\neg\mathcal{K}}(w)}, \qquad \Pi(g) = \frac{\sum_{w \in g} E_1^{\mathcal{K}}(w)}{\sum_{w \in g} E_1(w)}.$$

Then, splitting the numerator of R_{eff} and multiplying each term by 1,

$$R_{\text{eff}}(g) = \frac{\sum_g E_2^{\mathcal{K}} + \sum_g E_2^{\neg\mathcal{K}}}{\sum_g E_1} = \frac{\sum_g E_1^{\mathcal{K}}}{\sum_g E_1} \cdot \frac{\sum_g E_2^{\mathcal{K}}}{\sum_g E_1^{\mathcal{K}}} + \frac{\sum_g E_1^{\neg\mathcal{K}}}{\sum_g E_1} \cdot \frac{\sum_g E_2^{\neg\mathcal{K}}}{\sum_g E_1^{\neg\mathcal{K}}} = \Pi(g)\, R_{\text{CpG}}(g) + (1 - \Pi(g))\, R_{\text{non}}(g),$$

using only that expected counts add and that $\sum_g E_1^{\neg\mathcal{K}} / \sum_g E_1 = 1 - \Pi(g)$. This is an **exact identity**, not a fit, so the panel reads additively.

Reading the panel. Its four curves are these symbols, in this order: R_{eff} (solid orange diamonds, what Gnocchi applies), R_{non} (black dashed squares) and R_{CpG} (black dotted triangles) — the two terms it decomposes into — and the counterfactual derived next, drawn in R_{eff}'s own colour and symbol but dashed and hollow, because it *is* R_{eff} with one term switched off. The y-axis is $R(g)$ itself, a ratio of summed expected counts, so 1 means *no adjustment* and the horizontal line at 1 is the null. Since R_{eff} is a Π-weighted average of the other two, it must lie between them: it tracks R_{non} closely at low GC, where Π is small, and is pulled down towards $R_{\text{CpG}} \approx 1$ as Π grows.

The counterfactual, and what it intervenes on. The question is how much of R_{eff}'s rise the CpG contexts could account for on their own. So switch off the *non-CpG* adjustment — set $r_t \equiv 1$ for $t \notin \mathcal{K}$, which sends $E_2^{\neg\mathcal{K}} \to E_1^{\neg\mathcal{K}}$ — and change nothing else: the fitted r_t for $t \in \mathcal{K}$ stay, the weights $\Pi(g)$ stay. The dashed orange curve is

$$R_{\text{eff}}\big|_{r_t\equiv 1,\ t\notin\mathcal{K}}(g) = \frac{\sum_g E_2^{\mathcal{K}} + \sum_g E_1^{\neg\mathcal{K}}}{\sum_g E_1} = \Pi(g)\, R_{\text{CpG}}(g) + (1 - \Pi(g)),$$

i.e. **what Gnocchi would apply if it adjusted CpG contexts alone**. It is flat within 0.4% across the whole GC range while R_{eff} climbs to 1.33, so *none* of the applied trend survives the removal of the non-CpG term: the GC dependence of what Gnocchi applies is wholly non-CpG.

Flatness here is a result, not an identity. Π reaches **0.26** in the highest bin the panel draws, so a GC trend in R_{CpG} would appear in this curve scaled by Π, not erased — the curve is flat because R_{CpG} itself is (0.997–1.014), which is the next paragraph's subject and is *correct* rather than a failure.

Where each quantity comes from. `data._r_eff_components` builds the four per-window sums, one SQL query, and `data.r_eff_by_gc` does the binning and the divisions:

symbol	column	built from
$E_1(w)$	`e1`	`expected_counts_by_context_methyl_genon` — published, already summed over all 32 contexts, $r \equiv 1$
$E_2(w)$	`e2`	`refits/expected_counts_by_context_methy` — the same windows after the refit's r
$E_1^{\mathcal{K}}(w)$	`e1_cpg`	`expected_counts_per_context_methyl_genc` the per-(w,t) export, restricted to $t \in \mathcal{K}$ and summed
$E_2^{\mathcal{K}}(w)$	`e2_cpg`	the same rows times `rr` from `refits/rr_by_context.{pop}.txt`, the per-(w,t) adjustment
$E_1^{\neg\mathcal{K}}, E_2^{\neg\mathcal{K}}$	—	subtraction, `e1 - e1_cpg` and `e2 - e2_cpg`

The subtraction is why only the CpG slice of the two multi-GB per-context files is ever joined: an 85M × 85M join becomes 10M × 10M. It also means the identity above mixes two published files — totals from the summed export, CpG parts from the per-context one — so it needs them to describe the same counts. They do: `preconditions/verify_expected_r1.py` regenerates the first from the second genome-wide, `possible` exactly and `expected` to 4.6×10^{-5} relative.

R_{eff}, R_{CpG}, R_{non}, Π and the counterfactual are then the columns `r_eff`, `r_cpg`, `r_non`, `pi_cpg`, `r_counterfactual`. One further column, `r_eff_published` $= \sum_g E_2^{\text{pub}} / \sum_g E_1$, replaces the refit's numerator with Chen et al.'s own published `expected`. It needs no per-context r, which is what makes it a check rather than a restatement: the two agree to 1.7×10^{-5} per bin, median 2.1×10^{-6}, and that is what licenses using the refit's per-context r above — the published pipeline writes its own only to a local directory, never to the bucket.

Why $R_{\text{CpG}} \approx 1$ is correct, not a failure. CpG mutability is dominated by methylation, and p_c is already keyed by methylation level — across methylation 0 to 15 the CpG C>T rate spans **3.0–4.3×** depending on context, the largest single rate effect in the model, inside one trinucleotide. High-GC CpGs are CpG islands: **92% hypomethylated in the top GC bin**, against 2.5% in the GC bulk, and their empirical DNM rate falls from 0.532 in the bulk to 0.283 in that bin (**1.9×**). **Step 1 has already applied that correction**, via the covariation of GC content with methylation, so there is nothing left for r to adjust. (Chen et al. also strip `GC_content`, `CpG_island`, `Nucleosome`, `SINE` and `met_sperm` from CpG-context models, so those models could not express a GC dependence even if one were needed.) The GC trend in what Gnocchi applies is therefore entirely non-CpG — which is where the rest of this figure looks.

5 Panel C — the training set is not the scored population

r is fit on de novo mutations and matched background sites drawn from the **whole genome**, but applied to — and here evaluated on — the scored population. Panel C measures how far apart those two drift with GC.

Two scored populations, and the panel follows whichever is in force. `NEUTRAL_WINDOWS_BED` selects between the 1,843,559 windows this repo builds from the bucket (noncoding + `pass_qc` + autosome/PAR) and McHale et al.'s own 693,270. Either way the genome splits three ways — **QC-pass noncoding, QC-pass coding, QC-fail** — and unset, the first of those *is* the scored population, so the stack has three bands. Set, QC-pass noncoding subdivides into their set and the rest of it, giving a fourth:

$$N(g) = N_{\text{scored}}(g) + N_{\text{other}}(g) + N_{\text{coding}}(g) + N_{\text{QC}}(g).$$

Map each non-CpG training site — **both classes, DNMs and background alike** — to its containing 1 kb tile; the stack plots those fractions per GC bin, and they sum to 1 by construction. Nothing here re-derives a filter: a site is *scored* exactly when its window is a row of the analyzed window table, the same table `refit.py` restricts the training set with and panels A, B and E are evaluated on. So the bands track the configuration on their own, and a stratum that is empty draws no band and no legend entry. Between the two configurations QC-fail is identical — it is the windows with no row in Chen et al.'s table — and what moves is the QC-pass noncoding territory, cut into the scored and other bands. The coding band can shift a little too, by any window their file lists that this repo would call coding; the join prints that count.

The numbers below are from the narrowed run — `NEUTRAL_WINDOWS_BED` set, which is what this notebook's panels were built from. So the stack has four bands, its bottom one is McHale

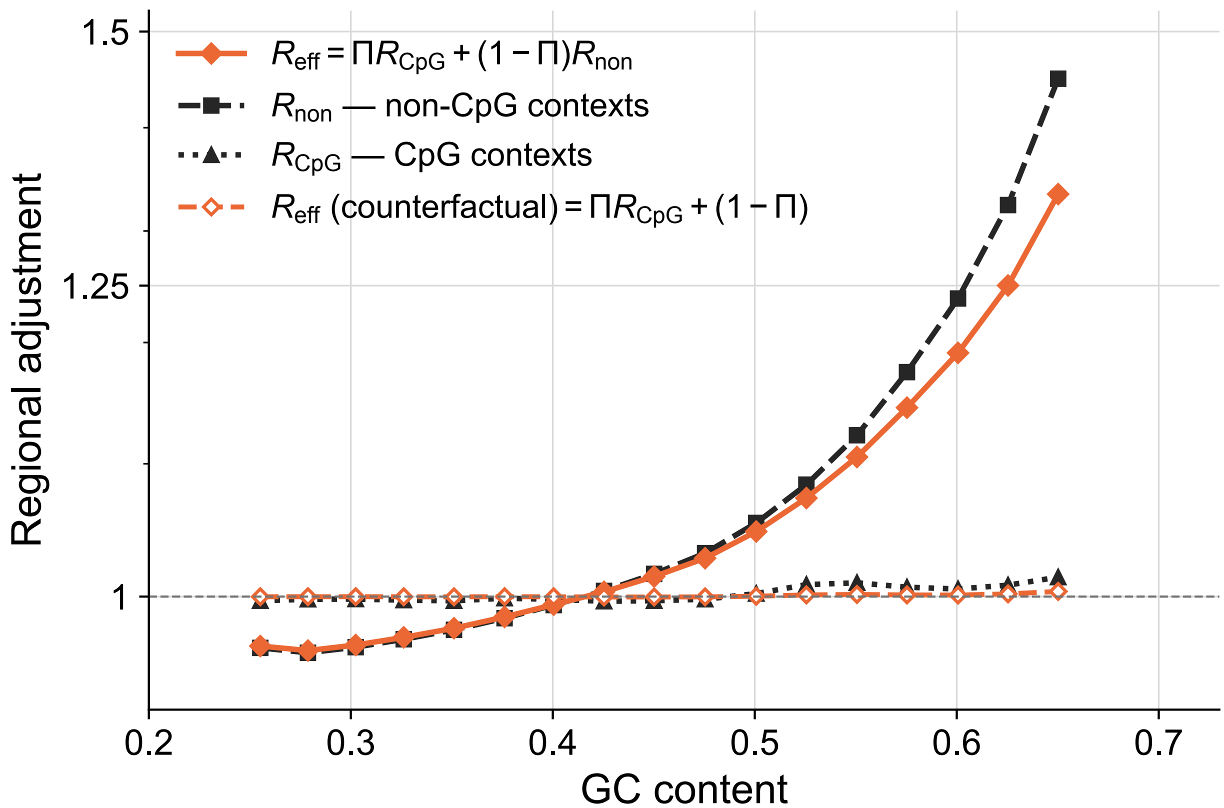


Fig. 5B | The GC dependence of the applied adjustment is wholly non-CpG. R_eff, the solid orange curve, is what Gnocchi applies. It decomposes exactly, as an algebraic identity rather than a fit, into R_eff = Pi*R_CpG + (1 - Pi)*R_non, the same ratio of summed expected counts restricted to the four CpG contexts and to the other 28, with Pi the CpG share of step-1 expected counts. R_non rises from 0.96 to 1.45 across GC while R_CpG stays within 0.997-1.014. The dashed orange curve, drawn in R_eff's own colour and symbol because it is that same quantity with the non-CpG term switched off, is the counterfactual Pi*R_CpG + (1 - Pi) – what Gnocchi would apply if it adjusted CpG contexts alone; it is flat within 0.4% while R_eff climbs to 1.33, so none of the applied trend survives removal of the non-CpG term. That flatness is a measurement, not an identity: Pi reaches 0.26 in the highest bin drawn (Supporting Fig. 7D), so a GC trend in R_CpG would appear here scaled by Pi rather than erased. R_CpG close to 1 is correct rather than a failure, because the large CpG methylation effect is already applied in step 1 (Supporting Fig. 7).

et al.'s 693,270 windows, and the lower row's ratios divide by that band rather than by QC-pass noncoding as a whole.

Why both classes. In a case-control design it is the controls that carry the covariate distribution, which argues for stacking the background class alone — but that is an argument about the *design*, not the *fit*. The fit minimizes its loss over the mixture, so the mixture is the training distribution, and that is what belongs beside the scored population. Stacking the background class alone barely moves the picture — on the 1,843,559-window reproduction, where this was measured, the QC-fail band runs 0.13 → 0.24 instead of 0.14 → 0.28 — because it outnumbers the DNMs about 12:1; what the DNM class adds is its own steeper drift, 64% QC-fail at GC 0.64 against the background class's 26%.

What the two rows are between them. The upper row is *covariate shift*: $P(x)$ in the training data is not $P(x)$ in the scored data, and the gap grows with GC. The lower row is *concept shift*:

$P(\mathrm{DNM} \mid x)$ is not the same function in the territory that drops out. Both are driven by one latent variable — which class of window a site sits in — and r cannot see it, being fit per trinucleotide context on regional features, none of which encodes window class. The drifting case-control ratio is not itself the problem: under case-control sampling the fitted slopes stay consistent whatever the ratio, and only the intercept shifts. Covariate shift alone would also be survivable, since a correctly specified model extrapolates. It is the combination — a mis-specified smooth fit, over a covariate range where the label-generating process is a different one — that puts a GC slope into r.

The bands above the scored one, where the labels matter because the mechanism differs. A *coding* window has a published Gnocchi score; it is this analysis that sets it aside, following McHale et al.'s noncoding restriction. A *QC-pass putatively nonneutral noncoding* window is QC-pass and noncoding but outside their set — the rest of that category, given up in narrowing to it, for an enhancer overlap or one of their interval exclusions. Its label is the complement of the bottom band's own, *QC-pass putatively neutral noncoding*, so the two read as one partition of that category rather than as a named class beside a leftover; *putatively* is load-bearing on both sides, because being outside a set McHale et al. call putatively neutral is not itself evidence of selection, and whether these windows differ at all is exactly what the lower row measures. A *QC-fail* window has no score at all — Chen et al. dropped it before scoring, keeping only windows with $\geq$ 1,000 possible variants, $\geq$ 80% of observed variants PASS, and mean coverage 25–35$\times$ (`run_nc_constraint_gnomad_v31_main.py:296`). Since `pass_qc` is `True` on all 1,984,900 rows of the published table, that flag is inert and a QC failure shows up only as an absent row.

That last band is therefore **not** "no gnomAD coverage", the name it carried here until measured: of the 587,902 absent autosomal windows, all have their QC inputs on file, and the PASS rule dominates the failures (417,097, against 19,396 for the coverage band; `preconditions/verify_qc_filter.py` has the full split, and confirms the filter forwards too — all 1,984,900 scored windows satisfy all three conditions). It is also a **mixture** of coding and noncoding, deliberately: 6.9% of these windows overlap coding exons against 7.1% of the QC-pass ones, so QC failure is near-independent of coding status and this is not a coding band in disguise.

The QC-pass noncoding share of the training set falls from 0.82 in the GC bulk to 0.27 by GC 0.68, and the scored band — the part of that territory McHale et al. call putatively neutral — peaks at 0.36 near GC 0.35 and is down to 0.007 over the same distance. So the model is fit on one population and applied to another, and the two come apart exactly where panel A's bias is largest.

The part of the DNM training set that lies outside the scored population has a different DNM rate — that is what the lower row measures, and the difference is not mainly the coding part. Writing $\bar{y}_s(g)$ for the empirical non-CpG DNM rate in stratum s and bin g, it plots $\log[\bar{y}_s(g)/\bar{y}_{\text{scored}}(g)]$ for each excluded stratum, against a linear axis, with error bars $\pm\,$SE where

$$\mathrm{SE}\Big(\log\frac{\bar{y}_s}{\bar{y}_{\text{scored}}}\Big) = \sqrt{\frac{1-\bar{y}_s}{k_s} + \frac{1-\bar{y}_{\text{scored}}}{k_{\text{scored}}}},$$

the delta-method binomial SE with k the DNM count. That SE is symmetric in the log ratio and in nothing else, so the log is the scale on which the bar drawn is the interval the data support; it also puts a 2$\times$ excess and a 2$\times$ deficit equally far from the reference line. Zero would mean those sites are exchangeable with the scored ones as far as mutation rate goes. The coding stratum sits there — 0.86–1.00$\times$, i.e. $|\log| \lesssim 0.15$, flat across the whole range — while the QC-failing stratum runs 1.50–1.63$\times$ (log = 0.41–0.49) through the GC bulk and **3.39$\times$ (log = 1.22) by GC 0.58**.

Under the narrowed population the curve to read first is the putatively nonneutral one, because it is precisely what the narrowing gave up: it is flat, at 0.94–1.03×, so the narrowing costs sample size and nothing else and the rest of this figure carries over unchanged. Had it climbed with GC it would have been a third population change, and would have belonged in the caption beside the other two.

Both rows come from one table of per-stratum counts — this row divides the DNM count by the site count, the row above takes the site count as a share of its bin. 72,801 of the non-CpG autosomal DNMs fall in the QC-failing stratum and 17,537 in coding windows, against 241,487 in QC-pass noncoding ones — 91,906 of those in the scored band, 149,581 in the putatively nonneutral rest of it.

So the steep GC dependence the model learns comes from sequence gnomAD could not call reliably, which is also where trio DNM calling is least reliable: part of that excess is plausibly false-positive DNM calls rather than real mutation.

6 Panel D — restricting the training set flattens the DNM rate, and the fit can then track it

What the two curves of a pair are estimating — one number, by two routes. Fix a training population and work over its non-CpG sites only. Write S_g for the sites whose GC content falls in bin g, $y_i \in \{0,1\}$ for site i's DNM label, and x_i for its feature vector. The quantity the panel is about is the probability that a training site in that bin carries a DNM,

$$P_g \;\equiv\; \Pr(y=1 \,\big|\, \mathrm{GC}\in g) \;=\; \mathbb{E}\Big[\,\Pr(y=1 \mid x)\,\Big|\,\mathrm{GC}\in g\Big],$$

the second form by the tower property — and its inner conditional is exactly what each trinucleotide context's logistic model estimates. The panel plots one estimate of P_g from each form, both plug-in averages over the sites the bin actually holds:

$$\bar{y}(g) = \frac{1}{|S_g|}\sum_{i\in S_g} y_i \;\approx\; P_g \quad \text{(empirical)}, \qquad \hat{p}(g) = \frac{1}{|S_g|}\sum_{i\in S_g} \hat{\pi}_i \;\approx\; P_g \quad \text{(fitted)},$$

where $\hat{\pi}_i = \widehat{\Pr}(y=1 \mid x_i)$ is the site's own fitted probability from its own context's model, evaluated at the **site's** feature vector rather than at the window-aggregated values the genome-wide apply uses.

The left one is the binomial MLE of P_g: unbiased, with standard error $\sqrt{\bar{y}(1-\bar{y})/|S_g|}$, which is what the error bars carry. The right one is a Monte-Carlo average of the fitted conditional over the same sites, so it estimates the same P_g **provided the model is calibrated on that bin's covariate distribution** — and departs from it when the model is not, which is the whole point of the panel. So $\hat{p}(g) - \bar{y}(g)$ is a calibration gap conditioned on GC, and each pair is a reliability diagram on its own population: *fitted vs empirical within a pair* is the comparison that means something.

Two cautions on what P_g is *not*. It is a probability under the **training** distribution, not the genome's: every DNM is kept while background sites are sampled at some rate s, which multiplies

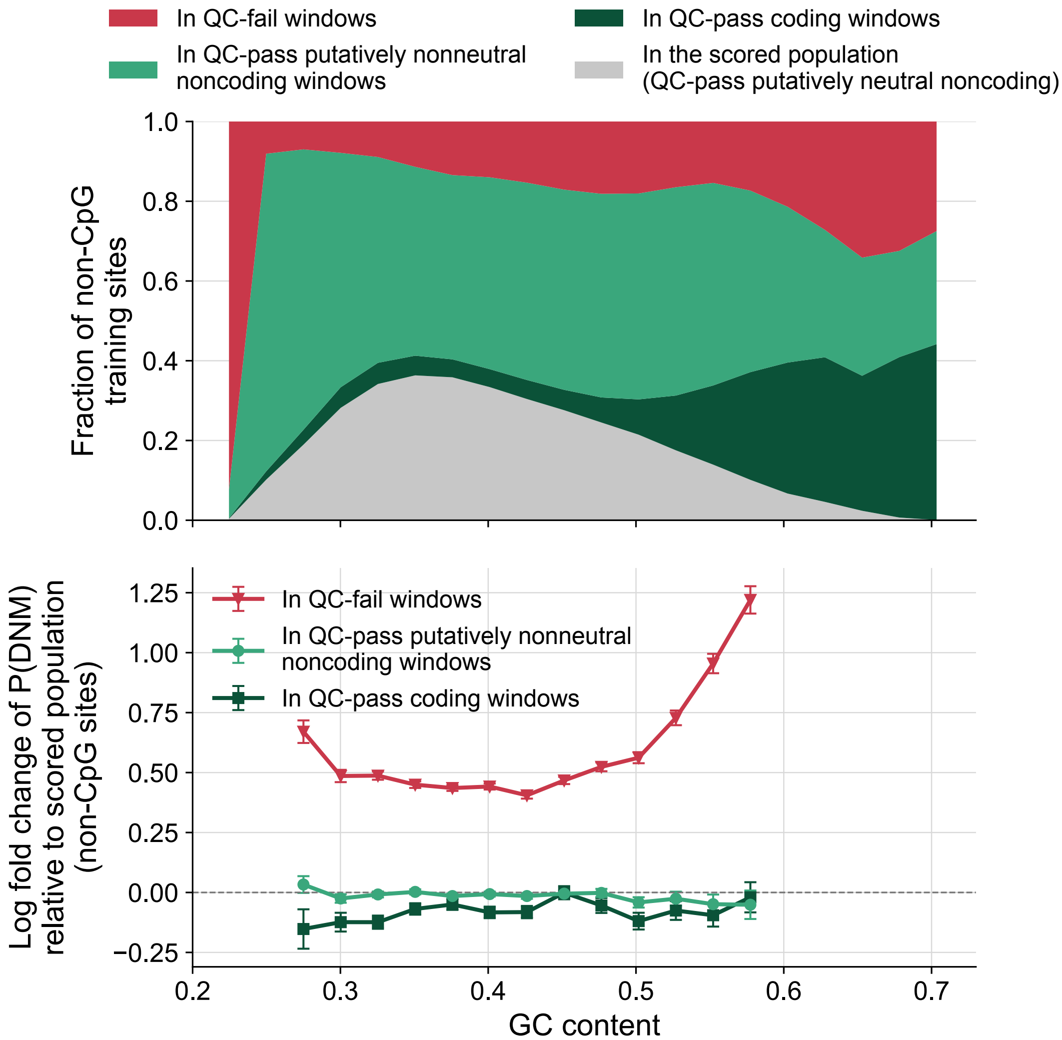


Fig. 5C | The training set is not the scored population, and the territory that drops out is different rather than merely absent. Upper, the composition of the 4,367,225 non-CpG training sites per GC bin, each mapped to its containing 1 kb window and stacked by that window's stratum, so the four fractions sum to 1 in every bin. The scored share peaks at 0.36 near GC 0.35 and falls to 0.02 by GC 0.65, while the QC-fail share rises from about 0.15 in the GC bulk to 0.34. Lower, the log of each excluded stratum's empirical non-CpG DNM rate divided by the scored population's in the same bin, with error bars of +/- 1 delta-method binomial s.e.; zero means those sites are exchangeable with the scored ones as far as mutation rate is concerned. The coding stratum sits there (0.86-1.00x, flat across the range) and so does the QC-pass putatively nonneutral noncoding stratum (0.94-1.03x), so narrowing to the putatively neutral set costs sample size and nothing else. The QC-failing stratum runs 1.50-1.63x through the GC bulk and 3.39x by GC 0.58. The model is therefore fit on one population and applied to another, and the two come apart exactly where the bias in (A) is largest. That excess lies in sequence gnomAD could not call reliably, which is also where trio DNM calling is least reliable, so part of it is plausibly false-positive DNM calls rather than real mutation.

the odds by $1/s$ and so shifts logit P_g by $-\log s$ — a constant in x, and in particular in GC. Levels are therefore not comparable **across** populations, whose case-control ratios differ (12.2, 12.2 and 13.4 background sites per DNM), while shapes in g are; each curve is divided by its own site-weighted mean, leaving shape only. And g enters only as a conditioning event: GC is one of the 13 candidate regional features, but $\hat{p}(g)$ averages whatever the fit selected for that context, so a pair can come apart in a bin even where GC itself was never selected.

Two populations:

- **original** — the training set as published;
- **scored** — restricted to the analyzed windows (the intervention).

A third, **size-matched** — the same *number* of sites as *scored*, drawn at random from the whole genome — is fit and reported, but not drawn here. It is the control that separates "better-matched population" from "less data", and it lies on top of the original pair; a curve indistinguishable from one already on the axes costs two of this panel's series to say nothing the number cannot. It is reported under panel E, where the same control is measured on the statistic that matters.

Two things happen at once when the training set is restricted. The **empirical** GC dependence itself shrinks and becomes monotone — on the original set $P(\text{DNM})$ rises 2.45× and then *collapses* above GC 0.66; on the scored set it rises smoothly by 1.60× with no turnover. And the logistic regression can then actually fit it: on the original set it misses by 26% and 29% *in opposite directions*, because it is a smooth monotone surface chasing a curve that turns over, where on the scored set it tracks to within 6% through GC 0.58. The scored pair's final bin is the exception — 670 sites, and the fit 28% low there — so the claim is about the GC range the restriction leaves populated, not about the tail it thins.

This is why R_{non} in panel B is inflated — the model has partly learned a GC slope that belongs to sequence outside the scored population.

Caveat. This panel measures a *level* error in $P(\text{DNM})$, and levels cancel in r. It diagnoses the fit; it is not itself a measurement of Gnocchi's bias. Panel E is.

7 Panel E — refitting r on the scored population removes Gnocchi's bias

Exactly one thing changes relative to the published pipeline: training sites outside the analyzed window set are dropped. Everything downstream is identical — univariate Bonferroni selection, standardization, `IncrementalPCA`, L1 logit per context, genome-wide apply. So panel E is panel A's statistic with a third expected count, E_2^{scored}, added to the same joint z filter and the same ranking.

Two controls make this a result rather than an anecdote, and both are printed below rather than plotted (a curve indistinguishable from published Gnocchi is clutter):

- the **full-population refit** through this same code lands on published Gnocchi, so "before" and "after" differ by the intervention and not by whose code produced them;
- the **size-matched random refit** also lands on published Gnocchi, so the improvement is about *which* sites, not *how many*.

Note the retrained curve is expected to end up *better than* the context-only model, not merely

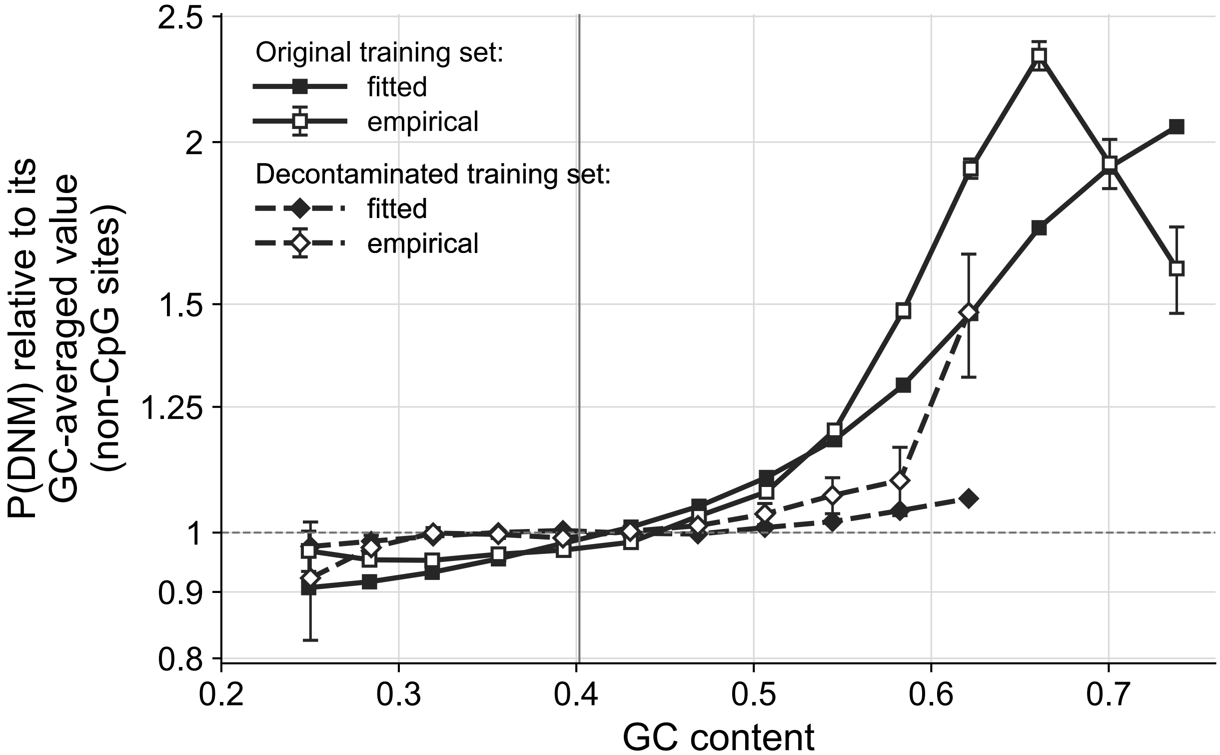


Fig. 5D | Restricting the training set flattens the empirical DNM rate, and the fit can then track it. Fitted and empirical P(DNM | GC bin) for two training populations, each population drawn as a pair sharing one symbol and one line style (squares and a solid line for the original training set, diamonds and a dashed line for the decontaminated one) and legended as one group; the empirical member of each pair is the one with open symbols and error bars. Each curve is divided by its own site-weighted mean across bins so that only shape is compared; the gap within a pair is a calibration gap conditioned on GC. On the training set as published (4,355,429 sites) the empirical rate rises 2.45x and then collapses above GC 0.66, and the fit, a smooth monotone surface chasing a curve that turns over, misses by 26% and 29% in opposite directions. Restricted to the scored population (930,417 sites) it rises smoothly by 1.60x with no turnover and the fit tracks it to within 6% through GC 0.58; the final bin, holding 670 sites, is the exception. A third training population, a size-matched random subsample drawn from the whole genome, lies on top of the original pair and is reported under (E) rather than drawn here. This panel diagnoses the fit: the level error it measures cancels in the ratio r, so (E) rather than (D) is the measurement of Gnocchi's bias.

closer to it: a correct r should repair the context-only model's own droop at both GC extremes. That is the sense in which this is a positive result and not just the removal of a defect. On this run it does: mean $|\rho - 0.5|$ is 0.168 for published Gnocchi, 0.046 for the context-only model and **0.026** for the retrained one, against 0.168 for the full-population control and 0.162 for the size-matched one. The first three of those five are the curves the panel draws, and it carries them in its legend; the two controls are print-only.

8 Supporting figure (Supporting Figure 7 in the manuscript) — why $R_{\mathrm{CpG}} \approx 1$ is correct

Panel B asserts that CpG contexts *should* apply no GC-dependent adjustment, because the effect that would need adjusting has already been applied in step 1. That claim rests on three measure-

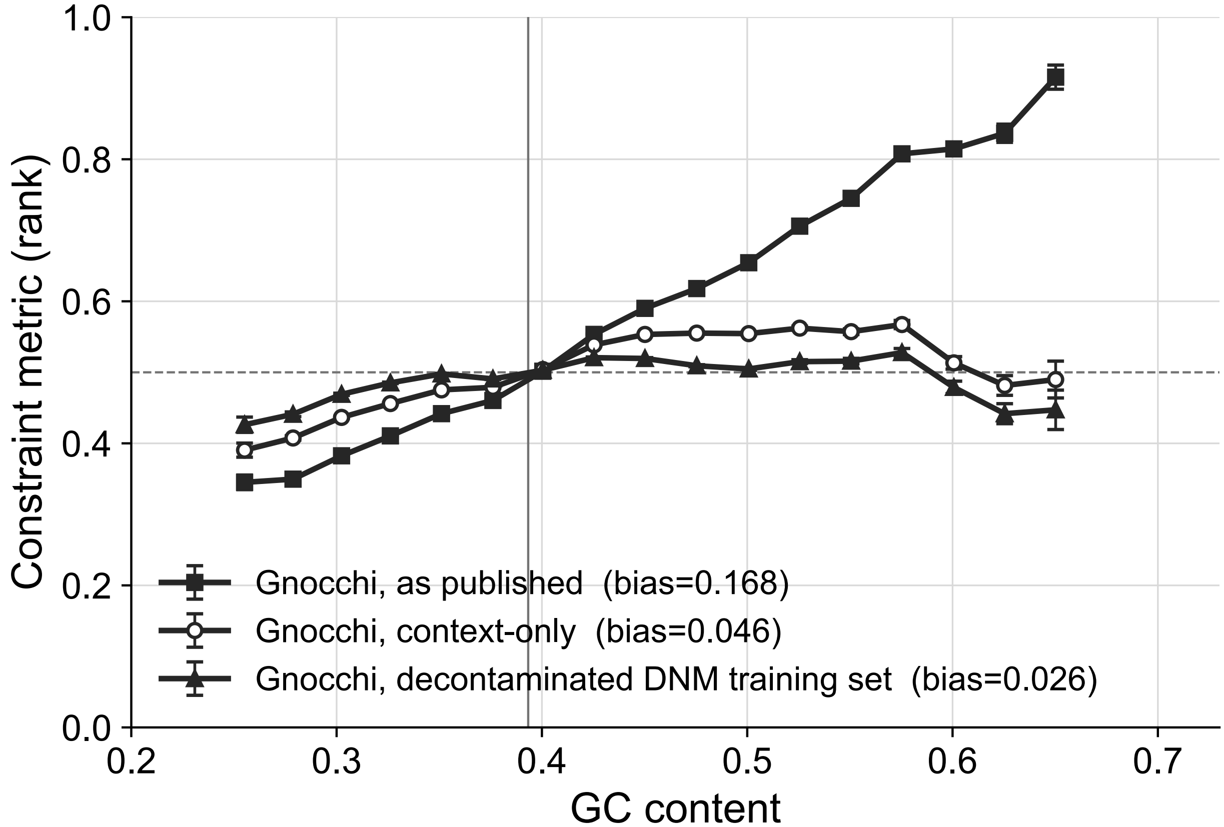


Fig. 5E | Refitting r on the scored population removes the bias. The statistic and the windows of (A), with a third score added, on the 692,325 windows surviving the joint filter across all three models. Exactly one thing changes from the published pipeline: training sites whose window lies outside the analysed set are dropped. Each curve's legend entry carries its bias. The retrained score falls below not only published Gnocchi but the context-only model itself, so a corrected r also repairs that model's droop at both GC extremes rather than merely undoing the damage. Two controls are reported numerically rather than plotted, each being indistinguishable from published Gnocchi: refitting through the same code on the full training set gives 0.168, so before and after differ by the intervention and not by whose code produced them, and refitting on a size-matched random subsample of the same training population (1,202,413 sites) gives 0.162, so the improvement is about which sites, not how many.

ments which the panel itself does not show, so they get their own figure, `output/supp_fig7.pdf`:

A. The size of the methylation effect step 1 absorbs. p_c is keyed by methylation level, and for CpG C>T the fitted probability spans **3.0–4.3×** from level 0 to 15 — within a *single* trinucleotide context, and the largest single rate effect in the model. The dashed curves are `mu`, the same observable read in a 1,000-genome cohort and rescaled, which spans 9.7–15.2× over the same range; it is not itself a measured rate, and the subsection below says what it is. The gap between the curves is the saturation of p_c.

B. High-GC CpGs are CpG islands. The hypomethylated fraction (level ≤ 1) runs 2.5% through the GC bulk and **92% in the top GC bin**, the one bin above GC 0.70; mean methylation falls from ~6.4 to 0.47 over the same range.

C. So their DNM rate collapses. Flat at 0.532 through the bulk, falling to **0.283** in the top

GC bin — a 1.9× fall, tracking B. The two highest bins hold 932 and 1,434 sites; their error bars carry that.

D. And these contexts are not a rounding error. Π, the CpG share of a bin's step-1 expected counts — the same weight panel B's identity $R_{\text{eff}} = \Pi R_{\text{CpG}} + (1-\Pi)R_{\text{non}}$ uses — rises from **0.038 to 0.264** across the GC range. So the flatness of panel B's counterfactual is a measurement and not an artifact of negligible weight: had R_{CpG} carried a GC trend, it would have reached the applied multiplier scaled by up to 0.26. This is the panel that says the A–C mechanism *matters* rather than merely *holds*.

Put together: a large, strongly GC-dependent CpG effect exists, step 1 already applies it, and the contexts it applies to carry up to 26% of the expected counts. There is nothing left for r to correct, which is exactly what panel B measures. (Level, not rate: these are case-control-sampled training sites at ~10:1, so the y-axis of C is not a genome-wide mutation rate.)

D is binned over **Chen windows** and A–C over **training sites**, the same split the main figure runs on, so the two do not end in the same place: D's last bin above the 100-window floor is centred at GC 0.65 against 0.73 for B and C. All four panels share this figure's 0.2–0.8 axis. On the narrowed window set D no longer runs past panel B's own 0.2–0.73 range; that it did — reaching GC 0.75 with Π at 0.43, so that the bins carrying the most CpG weight were exactly the ones the main panel could not show — was a feature of the 1,843,559-window run.

8.1 Panel A's two curves

The model both curves live in. Infinite sites plus a genealogy, with u_c for class c's *true* per-generation mutation rate. Trace the ancestry of the cohort's n chromosomes back to their common ancestor and add up every generation of transmission in that tree: L_n generations in all, the genealogy's **total branch length**. Each one is an independent chance for the site to mutate at rate u_c, so the number of mutations the site accumulates anywhere in the cohort's ancestry is Poisson with mean $u_c L_n$.

```
                              above the MRCA: NOT part of L_n - a mutation
                              up there is inherited by the entire cohort,
                              so the site looks monomorphic, never counted
                          <- MRCA

                            X     <- one mutation, at rate u_c per generation




               c1    c2  c3    c4     <- the cohort: n sequenced chromosomes
                         ^^    ^^
                         c3 and c4 carry the variant, c1 and c2 do not,
                         so the site is POLYMORPHIC in the cohort

  L_n = total length of every branch below the MRCA, in generations
        (here 4 tip branches + 2 internal ones)
```

Two facts turn the mutation count into the thing the data reports. First, a mutation on any branch of that tree is inherited by *some but not all* of the cohort's chromosomes — branches above the common ancestor are not part of L_n, and are exactly the ones that would be invisible — so it shows up as a variant in the cohort. Second, *infinite sites* says no site is ever hit twice, so no mutation reverts or lands on an allele already present, and none of them cancel. The site is therefore polymorphic exactly when the count is non-zero, which has probability $1 - \Pr(\text{count} = 0)$:

$$P_n(c) \;=\; 1 - e^{-u_c L_n}.$$

A rate and a polymorphism probability are therefore two points on one curve, separated only by L_n — all of the cohort-size dependence sits there, and all of the sequence biology in u_c. The single-genome case is $n = 2$, where $L_2 = 4N_e$ and $P_2 \approx 4N_e u_c$ is heterozygosity: $4 \cdot 2 \times 10^4 \cdot 1.2 \times 10^{-8} \approx 9.6 \times 10^{-4}$, the textbook human per-base value.

Two cohorts, one parameter (Peter McHale's reduction, and the cleanest way in). Read the formula at the two cohort sizes the panel uses. In the 1,000-genome downsample saturation is weak, so the exponential linearizes:

$$P_{2000}(c) \;\approx\; u_c L_{2000}.$$

Substitute $u_c = P_{2000}(c)/L_{2000}$ into $P_n(c) = 1 - e^{-u_c L_n}$ and the rate drops out entirely:

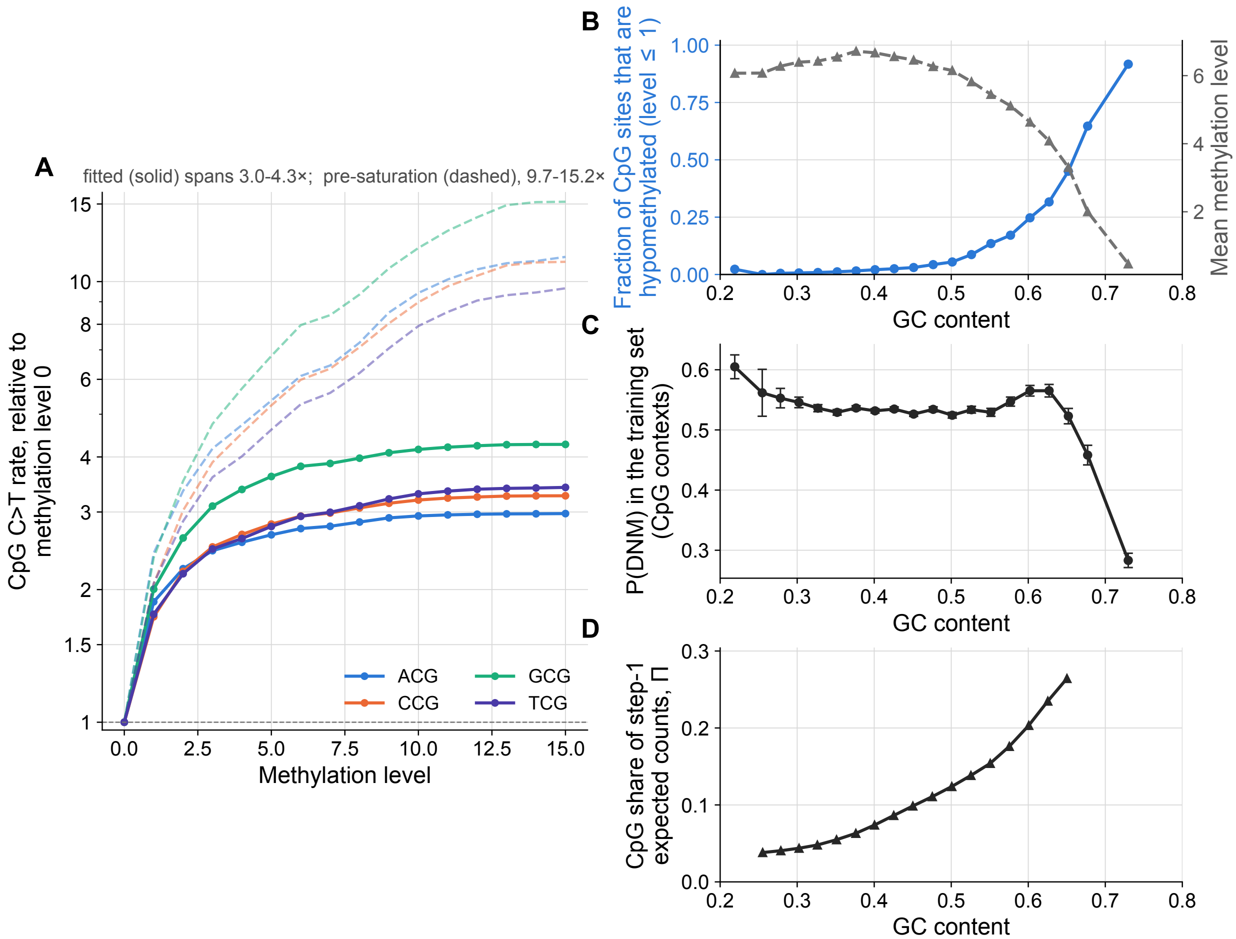
$$\boxed{P_n(c) \;\approx\; 1 - e^{-k\,P_{2000}(c)}, \qquad k = \frac{L_n}{L_{2000}}}$$

One parameter, and it is not a mutation rate: it is the **ratio of the two cohorts' genealogies**. Everything class-specific — context, methylation, the whole point of the table — has cancelled, and with it any need for a notion of an absolute mutation rate: the two cohorts calibrate each other.

The two curves are that formula read at the two cohort sizes. The **solid** curve is $P_n(c)$ itself at $n = 152{,}312$ chromosomes — p_c = `fitted_po`, the probability a site of that class is polymorphic in the 76,156-genome call set, and the operative output of step 1 (`expected = possible × fitted_po`). The **dashed** curve is Chen et al.'s `mu` column, which despite its name and its units is $P_{2000}(c)$ times one global constant; the panel divides each curve by its own methylation-0 value, so that constant cancels exactly and the dashed curve *is* P_{2000}, normalized. Saturated end and unsaturated end of the same relation, in other words — which is why the solid curve is so much the flatter of the two. Chen et al. plot the relation itself in their **Extended Data Fig. 1a**: proportion observed in 76,156 genomes against `mu`, "exponentially correlated" in their caption, fitted as $y = 1 - \exp(-1.88 \times 10^7 x - 7.32 \times 10^{-5})$ with $R^2 = 0.999$ — the boxed relation, in their units.

"Unsaturated" is relative. At 1,000 genomes the median row is 0.75% polymorphic, but the top of the dashed curve — ACG C>T at methylation 15 — is **27.6%**. Read as a Poisson rate, $-\log(1 - P_{2000})$, that context's span against methylation 0 is 13.2× rather than the 11.4× the raw proportion gives, so the dashed curve understates the true spread as well and the solid curve's compression is a lower bound.

What the gap does and does not imply. Saturation is not an error in Chen et al.'s model. The expected counts built from p_c are compared against observed *polymorphism* counts, which saturate identically, so it cancels — the same cancellation that makes a level error invisible in r.



Supporting Fig. 7 | Fig. 5B shows R_CpG flat at about 1 across the whole GC range. This figure shows that this is correct rather than a failure of the model: the large, strongly GC-dependent CpG effect that a regional adjustment might have been expected to carry is already applied in step 1, and the contexts it applies to carry enough of the expected counts for its absence from step 2 to be informative. See Methods, "How Gnocchi's regional adjustment drives its GC bias". (A) The size of the methylation effect that step 1 absorbs. Solid, the fitted C>T probability for each of the four CpG contexts against methylation level, each divided by its own value at level 0: it spans 3.0-4.3x from level 0 to level 15, within a single trinucleotide context and the largest single rate effect in the model. Dashed, the same observable read in a 1,000-genome downsample and normalised the same way, spanning 9.7-15.2x. The gap between them is saturation of the polymorphism probability at the larger cohort size, which cancels against the observed counts and is not an error in the model. (B) High-GC CpGs are CpG islands. Fraction of CpG-context training sites that are hypomethylated (methylation level <= 1; blue, left axis) and mean methylation level (grey, right axis), per GC bin. The hypomethylated fraction runs 2.5% through the GC bulk and reaches 0.92 in the top bin, while mean methylation falls from about 6.4 to 0.47 over the same range. (C) So their DNM rate collapses. Empirical P(DNM) at CpG-context training sites per GC bin: flat at 0.532 through the GC bulk and falling to 0.283 in the top bin, a 1.9-fold fall that tracks (B). Error bars +/- 1 s.e.m. These are case-control-sampled training sites, so the y axis is a probability under the training distribution and not a genome-wide mutation rate. (D) And these contexts are not a rounding error. Pi, the CpG share of a GC bin's step-1 expected counts and the weight that appears in Fig. 5B's decomposition, rises from 0.038 to 0.264 across the range. Had R_CpG carried a GC trend, it would have reached the applied multiplier scaled by up to 0.26, so the flatness of Fig. 5B's counterfactual is a measurement and not an artefact of negligible weight. (A)-(C) are binned over training sites and (D) over the analysed windows, so the two do not end in the same GC bin. All four panels share the same 0.2-0.8 axis.

9 Values quoted in the figure captions

Every number the captions quote is computed here.

```
panel A: 692,348 windows after joint z filtering, mean GC 0.393
  mean |rank - 0.5|  context-only (r = 1)              0.046
  mean |rank - 0.5|  Gnocchi as published              0.168
  mean |rank - 0.5|  depletion rank                    0.096

panel B: r_non spans 0.960-1.450; r_CpG spans 0.997-1.014; counterfactual spans
1.000-1.004
  CpG share of expected counts rises 0.038 -> 0.264

panel C: 4,367,225 non-CpG training sites in the plotted range; scored-
population fraction 0.00 at GC 0.22 -> 0.00 at GC 0.70

panel D: empirical P(DNM), max / min ratio within each population
  full         2.45x   (fitted 2.27x)
  scored       1.60x   (fitted 1.09x)

panel E: 692,325 windows after joint z filtering
  mean |rank - 0.5|  context-only (r = 1)                        0.046
  mean |rank - 0.5|  Gnocchi as published                        0.168
  mean |rank - 0.5|  Gnocchi, retrained on the filtered training set 0.026
  mean |rank - 0.5|    control: refit on the full training set    0.168
  mean |rank - 0.5|    control: size-matched random subsample     0.162
```

10 Caveats

- **Panel D is in-sample.** The retrained model is fit on training sites in the analyzed windows and scored against the labels of those same sites. Panel E is the out-of-sample confirmation: it is measured on gnomAD polymorphism counts, which the DNM model never sees. A held-out DNM split would make panel D airtight; it has not been run.
- **The depletion-rank curve comes from a different window set** (Halldorsson windows, a different window size) than the two Gnocchi curves. They are not joined; each is ranked within itself, and each is binned on its own GC edges. Halldorsson's windows also **overlap**, so no error bar is drawn for that curve: the within-bin windows are not independent, which the shared `std / sqrt(n)` assumes. It reads last in the legend for the same reason — it is the external comparison, not a third Gnocchi curve.
- **This run used McHale et al.'s 693,270 windows**, and the caption says so. `NEUTRAL_WINDOWS_BED` unset would mean the 1,843,559 noncoding + `pass_qc` + autosome/PAR windows this repo builds from the bucket; set, as here, it means their 693,270, taken from their file with none of those three filters applied on top — theirs is the definition, so it enters whole (`windows.build_window_table`). Either way it is one definition applied consistently: every panel, the population the retrained model is fit on, and panel C's bottom band, which is membership in that population rather than a re-derivation of its filters. The gap between the two is 2.66x and is not only enhancer-overlapping windows — their assembly-gap, ENCODE-exclude and low-coverage exclusions are in it too — so a result that

holds on only one of them is a result about the window definition.